Empowering Developing Nations: Sustainable Launch Approaches and Emerging Technologies for Small Satellites

A.M Asik Ifthaker Hamim[1], Abdullah Al Abrar Chowdhury[2], Sazzad Hossen Chowdhury[3], Amad Uddin Osama[4], Anik Chowdhury[5], Mohammed Moin Uddin[6] and Rashedul Arefin Ifty[7]

[1,3,4,5,6,7]Department of Computer Science and Engineering, International Islamic University Chittagong Kumira, Chattogram-4318, Bangladesh

[2]Department of Information Technology, American National University

1813 E Main St, Salem, VA 24153, USA

Email- [1]asikifthakerhamim75@gmail.com, [7]r.a.ifty2001@gmail.com,

Abstract—As humanity ventures deeper into the cosmos, the demand for innovative and sustainable solutions to address the complexities of space exploration continues to grow. This study investigates a wide range of strategies and technologies aimed at overcoming the challenges faced by small satellites, which are pivotal to this evolving landscape. Leveraging immersive missionspanning visualization, the research identifies opportunities to optimize mission planning and readiness. Small satellite constellations are presented as an effective approach to enhance coverage, reliability, and cost efficiency. Furthermore, the potential of utilizing the lunar atmosphere is explored as a means to reduce launch mass and support sustainability. A philosophical perspective on the evolution of future space networks further enriches the discourse, providing insight into the long-term dynamics of space infrastructure. Amidst the rapid expansion of the small satellite industry, the study underscores the critical importance of collaboration and innovation in addressing these challenges, fostering a resilient and responsible space ecosystem, particularly for developing nations.

Index Terms—Small Satellite Constellations, Sustainable Space Technologies, Lunar Atmosphere Utilization, Space Ecosystem Development

I. Introduction

The small satellite industry is experiencing rapid growth, with the global market expected to reach USD 5.88 billion by 2030, driven by a 7.9% annual growth rate. This growth is attributed to various factors, including increased technical innovation, lower launch costs, and the development of automation technologies. Key players in the industry include Airbus, Lockheed Martin, and others. Small satellites, which are miniaturized and weigh less than 1000 kg, serve various purposes such as communication, surveillance, earth observation, and in-orbit inspection. They are utilized by commercial enterprises, non-profit groups, and educational institutions for conducting missions in low-Earth orbit. The advantages of small satellites include their lower cost, shorter development time, and the ability to be launched using smaller and less expensive launch vehicles [1]. Despite the growth of the small satellite industry, the short lifespan of small satellites

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remains a challenge [2]. To address launch challenges and ensure the long-term success and sustainability of the industry, advancements in technology are crucial. This includes the development of orbital maneuvering vehicles and laser communications, which can improve mission success rates, launch costs, and space sustainability [1]. In addition to technological advancements, policy and regulatory coordination are needed to create a supportive industry environment. This coordination can help streamline the launch process, ensure space sustainability, and enable new applications for small satellites. In summary, addressing launch challenges is essential for the small satellite industry's long-term success and sustainability. The industry's growth, driven by advancements in technology and a supportive regulatory environment, will continue to enable new applications and drive innovation in the space sector.

II. Insights of Small Satellites

A. Definition and Classification of Small Satellites

Small satellites, also known as miniaturized satellites or smallsats, are satellites of low mass and size, typically under 1,200 kg (2,600 lb). They are classified based on their mass and size, and different classification systems are used to categorize them [3]. The main classification groups for small satellites are:

1) Large Satellites: More than 1,000 kg 2) Medium-sized Satellites: 500-1,000 kg 3) Small Satellites:
   a) Minisatellite: 100-500 kg
   b) Microsatellite: 10-100 kg

c) Nanosatellite: 1-10 kg
d) Picosatellite: 0.01-1 kg
e) Femtosatellite: 0.001-0.01 kg

In addition to these classifications, there is a specific type of small satellite called a CubeSat. CubeSats are a class of nanosatellites that use a standard size and form factor. The

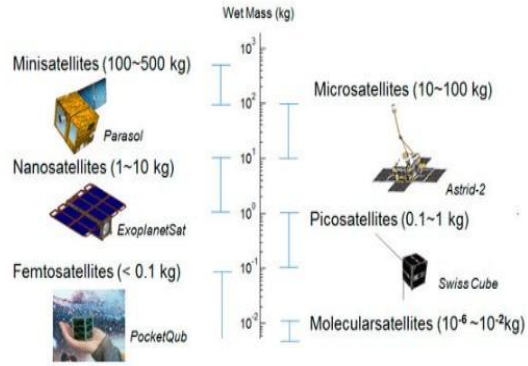

Fig. 1: Small Satellites Wet Mass

standard CubeSat size is a ”one unit” or ”1U” measuring 10x10x10 cm, but they can also be extended to larger sizes such as 1.5U, 2U, 3U, 6U, and even 12U. CubeSats were originally developed in 1999 to provide a platform for education and space research, and their standardization has led to cost-effective engineering and development projects.

B. Historical context of small satellite launches

The evolution of small satellite launches highlights technological progress, cost-efficiency, and standardized platforms. Sputnik 1 in 1957 marked the space age, showcasing small satellites' potential. Vostok 1 in 1961 spurred further innovation. By the 1970s, U.S. launch vehicles like Scout and Delta offered affordable deployment.

The 1990s brought the CubeSat standard, revolutionizing the industry for education and research. The 2000s saw dedicated launchers like Pegasus, Falcon 1, and Electron, enhancing flexibility and cost-effectiveness.

In the 2020s, small satellites surged for communication, Earth observation, and tech demonstration. SpaceX's Falcon 9 expanded opportunities by carrying multiple satellites. Driven by innovation, standardization, and demand, the industry now faces challenges in launch efficiency and space sustainability for future growth [4].

C. The increasing demand for small satellite missions

The small satellite industry [5] is rapidly expanding and is projected to grow further. Valued at USD 3.23 billion in 2022, the global market is expected to reach USD 5.88 billion by 2030, with a CAGR of 7.9%. The market spans North America, Europe, Asia-Pacific, and LAMEA. Key players include Airbus S.A.S., GomSpace, L3Harris Technologies, Lockheed Martin, Northrop Grumman, Planet Labs, Sierra Nevada Corporation, Thales Group, Aerospace Corporation, and Boeing.

Segmented by type, end-user, and geography, small satellites are used for communication, surveillance, Earth observation, in-orbit inspection, and more. However, their short lifespan poses a challenge to market growth [6].

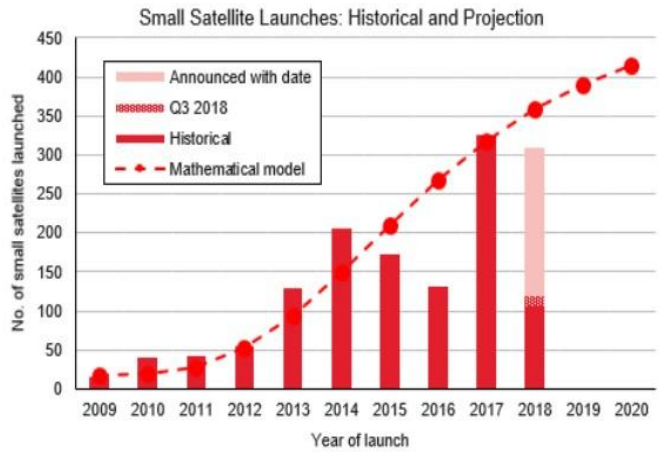

Fig. 2: Small Satellite Launches Historical and Projection Statistics

D. Challenges associated with launching small satellites

The rising demand for small satellite missions in developing countries stems from the growth of satellite-based communication, cost-effective solutions, and technological advancements. The top three challenges for small satellite launches are:

1) Space launch and space operations: Launching and operating small satellites pose challenges due to their size, mass, and limited kinematic capabilities. Managing failure risks is critical for small satellite programs [2].
2) Space sustainability and space safety: The increasing number of small satellites raises concerns about space debris and collision risks. Ensuring sustainability and safety requires effective space traffic management, debris mitigation, and adherence to international regulations [1].
3) Cybersecurity and supply chain: Small satellites face cybersecurity threats, necessitating robust system and data protection. Managing supply chains for components also poses challenges, including risks from counterfeit or compromised parts [7].

This research addresses space sustainability by proposing satellite classifications based on functionality rather than size or mass, aiming to minimize risks in higher orbits without hindering technical or economic potential [3].

III. Various Technologies Utilized for Small-Scale Space Endeavors

A. Immersive Mission Spanning Visualization

Immersive visualization technologies, blending physical and digital realities, create engaging, multi-sensory experiences. Immersive Mission-Spanning Visualization (IMSV), leveraging AR and VR, helps experts analyze and present complex data, addressing key challenges in small satellite missions [8].

By applying Immersive Mission-Spanning Visualization to small satellite missions, the following benefits can be achieved:

- Space sustainability and space safety: IMSV enhances space traffic management and debris mitigation by visualizing orbital environments and collision risks, aiding operators in minimizing debris and ensuring mission sustainability [1].
- Cybersecurity and supply chain: IMSV identifies system vulnerabilities, enabling robust cybersecurity measures. It also visualizes supply chains, helping detect counterfeit or compromised components [8].
- Access to space and launch delays: IMSV optimizes launch planning by providing real-time, interactive views of schedules and resources, reducing delays and costs [2].

It also offers solutions for the following challenges:

- Pre-launch Simulation: Tools like NASA's NOS3 and MATLAB validate spacecraft designs and reduce risks before launch [9].
- Training and Mission Control: VR simulations improve mission control readiness, while software like Open MCT and Core Flight System enhances decision-making. Immersive technologies also support psychological wellbeing during long missions and cloud-based planning [1]. • Public Engagement: IMSV fosters public involvement through interactive consultations, VR/360° videos, and virtual exhibits, making small satellite missions more accessible [3]

By tackling these challenges, IMSV drives the small satellite industry's growth, enabling efficient, cost-effective solutions for space exploration, communication, and Earth observation.

B. Satellite Constellations

Satellite constellations [10], comprising multiple satellites working together, offer advantages like redundancy, global coverage, improved performance, and reliability. If one satellite fails, others maintain network functionality, and distributed constellations are more resilient to attacks [11]. They provide global or near-global coverage, serving remote or dispersed areas effectively and enabling global connectivity and internet advancements.

Operating in lower Earth orbits (LEOs), constellations reduce signal delays, enabling faster communication and lower latency. Distributed data traffic enhances throughput, improving communication efficiency. For Earth observation, they capture high-resolution images and revisit locations frequently, aiding environmental monitoring, weather forecasting, and disaster management [12].

Constellations are scalable, allowing expansion or adjustment by adding satellites to meet changing demands. Mixed orbits (LEO, MEO, GEO) optimize performance, as seen in GPS. They also enhance data security through multiple communication paths and enable collaborative scientific research by collecting data simultaneously from various points.

Examples include Iridium [13] for satellite phones, Starlink for broadband, and Copernicus Sentinel for Earth observation. Cost reduction strategies include public-private partnerships, joint ventures, shared launches, and hosting third-party payloads. User contributions, such as pre-sales or subscriptions,

TABLE I: Description of Satellite Constellations

Constellation Name	Purpose	Intended Outcome
Planet Labs' Dove Constellation [14]	Daily revisit rates, enabling rapid Earth imaging for agriculture, forestry, and disaster monitoring applications	Capture valuable data at a fraction of the cost of traditional large satellites
Iridium NEXT Constellation [13]	Reliable and seamless coverage worldwide, even in remote regions	The feasibility of upgrading and modernizing satellite constellations over time
Globalstar LEO Satellite Constellation [14]	Expanded by launching additional satellites to meet increasing demand	Lower latency than geostationary satellites, making them suitable for applications like voice calls

can also fund development. Effective deployment aligns with project goals, whether focused on coverage, redundancy, reliability, or cost-sharing, making constellations more accessible for communication, Earth observation, and navigation [8].

These case studies highlight the benefits and challenges of small satellite constellations in Earth observation and global communication. Key lessons include the importance of cost efficiency, redundancy, scalability, and regulatory compliance. Future missions can build on these insights to enhance the capabilities and applications of small satellite constellations [10].

Satellite constellations also address the top three challenges of small satellite launches. By deploying fewer larger satellites instead of many smaller ones, they reduce collision risks and space debris [15]. On-board propulsion systems can further mitigate debris by de-orbiting satellites at end-of-life. Centralized supply chains lower risks of counterfeit or compromised parts, while advanced cybersecurity measures, such as encryption and secure protocols, protect system integrity and data.

Economies of scale reduce costs per satellite and improve launch success rates [11]. New technologies like reusable

rockets and air-launch systems enhance launch flexibility and reduce delays. By leveraging these advantages, satellite constellations offer a sustainable, cost-effective solution for small satellite operations, ensuring safety and scalability while addressing launch challenges.

C. Lunar Atmosphere Utilization

The lunar atmosphere, though nearly a vacuum, contains trace gases like helium, neon, argon, and volatiles such as water vapor and carbon dioxide. These resources, particularly water ice in shadowed craters, are vital for small satellite missions. Water can be split into hydrogen and oxygen via electrolysis, providing propellants for thrusters, enabling extended missions, orbit changes, or even Earth returns. Helium3, another lunar resource, holds potential for power generation, further enhancing small satellite capabilities [16].

Lunar resource utilization addresses the top three challenges of small satellite launches. NASA's Break the Ice Lunar Challenge focuses on excavating icy regolith to provide water resources, reducing reliance on Earth-based supplies and minimizing space debris [15]. Extracting oxygen from lunar soil for life support and propellant production reduces dependence on Earth-based supply chains, mitigating risks of counterfeit or compromised parts. Additionally, lunar power sources like solar panels or helium-3 fusion reactors offer sustainable energy, reducing the need for frequent launches and storage issues. By leveraging lunar resources, small satellite missions can achieve greater sustainability, security, and efficiency.

IV. Future Space Networks & Philosophical Perspective

Future space networks can be explored from a philosophical perspective to address some of the top problems in the field. Here are three key philosophical aspects that can help in solving these problems: Understanding the philosophical foundations of space and time can provide insights into the structure and dynamics of future space networks. For example, Kant's views on space as a subjective and ideal concept originating from the mind can be applied to the design and coordination of space-based systems. Similarly, the dynamic nature of being in process philosophy can help in conceptualizing the interactions and relationships within a future space network [4]. As future space networks incorporate artificial life, artificial intelligence, and artificial sociality, ethical considerations become crucial. Process philosophy offers a framework for investigating the exceptionality of human capacities and the scope of moral obligation in the context of these new phenomena. By applying ethical principles to the design and operation of space networks, we can ensure their responsible and sustainable development [8]. Future space networks require an integrated metaphysics that can join our common sense and scientific images of the world. This can be achieved by using category-theoretic tools to bridge the gap between philosophical concepts and scientific understanding. By developing a comprehensive framework that incorporates both metaphysical and scientific perspectives, we can address the complex challenges of future space networks.

By exploring these philosophical aspects, we can gain a deeper understanding of the problems and potential solutions in future space networks [16]. This interdisciplinary approach, combining philosophy, science, and technology, will be essential in shaping the future of space exploration and communication.

V. Advantages for small satellite launch problem resolution

Small satellites, such as CubeSats [17], have played a significant role in accelerating the growth of the small satellite industry, increasing access to space, and expanding low-Earth orbit research. Their smaller size and lower complexity allow for increased production cadence, enabling more frequent research missions and technology demonstrations. Building smaller, less expensive satellites reduces financial risk and lessens the probability of mission failure compared to larger, more complex satellites. The availability of commercial-offthe-shelf components and the shrinking of electronics and sensors have considerably reduced the cost of hardware development for small satellites. Small satellites can demonstrate new technologies, such as satellite network communications, that could reduce the need for ground stations, lower costs, and increase data downloads for satellite applications [2]. Integration with existing and emerging space networks can enhance their communication capabilities, coverage, and connectivity. The adaptation of small satellite technology has lowered the barrier to entry for developing nations without previous space capabilities, allowing them to make strides toward having a presence in low-Earth orbit. Small satellites can be configured for various missions, and their network within a constellation increases the functional options for directing proximity and focus of placement. Integration with existing and emerging space networks ensures their long-term relevance and scalability [8]. Small satellites have achieved near-parity with larger satellites in remote sensing and have been used for various scientific research purposes, such as astrophysics, heliophysics, and Earth science. Their smaller size, lower cost, and higher repeat rate make them suitable for a wide range of space science applications.

VI. Collaboration and Policy Considerations

A. The role of collaboration among space agencies, commercial launch providers, research institutions, private sector, national & international organization

Space exploration and usage require collaboration among various stakeholders, including space authorities, commercial launch service providers, research organizations, and the business sector [5]. Each nation approaches these issues differently based on its objectives, resources, and capabilities. For instance, NASA is the main space agency of the United States, focusing on planetary exploration, Earth science,

manned spaceflight, and the advancement of space technology. Russia's Roscosmos manages Russia's space program and collaborates with NASA and other space agencies on the International Space Station (ISS). The China National Space Administration's (CNSA) space capabilities have been growing quickly, with significant accomplishments in crewed spaceflight and lunar exploration [18]. The European Space Agency (ESA) carries out a variety of missions, such as planetary exploration, Earth monitoring, and scientific study, and works closely with NASA, Roscosmos, and other space organizations. The Indian Space Research Organization (ISRO) has become known for its efficient missions, such as the exploration of Mars and the moon, and works with foreign partners to launch satellites. Private enterprises like SpaceX and Blue Origin are also significantly influencing the exploration and commercialization of space. The top three challenges associated with launching small satellites are space sustainability and safety, cybersecurity and supply chain, and access to space and launch delays. Philosophically, satellite constellations can solve these problems by collaborating with governments or public agencies to share the costs of constellation development, partnering with other commercial entities to share the costs of satellite development, launch, and operation, offering opportunities for hosting payloads from third-party organizations on satellites, sharing launch costs by booking multiple satellites on the same vehicle, and involving end-users or customers in funding. Additionally, Lunar Atmosphere Utilization can provide solutions to these challenges by utilizing the resources available on the moon, such as water and minerals, to reduce the cost of satellite development and launch, and by establishing a permanent presence on the moon to reduce the need for launching satellites from Earth. Effective deployment strategies should align with the specific goals and requirements of the satellite constellation project, whether the focus is on coverage, redundancy, reliability, or cost-sharing, for the success of satellite constellations in various applications, such as communication, Earth observation, and navigation [4]. Collaboration among nations is essential for the growth of space exploration and usage. International cooperation in space exploration has the potential to reduce costs, broaden the scope of technical expertise, and promote educational activities throughout the region. The International Space Station (ISS) is an excellent example of international collaboration, with space agencies from the US, Russia, Europe, Japan, and Canada involved. International partnerships can also provide opportunities for joint space explorations and scientific discoveries, as well as personal relationships across international boundaries, building long-lasting trust and respect [8]. The Bangladesh Space Research and Remote Sensing Organization (SPARRSO) is the sole space research facility in Bangladesh. It is a state agency responsible for astronomical research and the application of space technology in the country. SPARRSO works closely with international partners such as JAXA, NASA, and the ESA in environmental and meteorological research. SPARRSO also conducts research in these areas to develop and apply space and remote sensing technology [19]. The organization serves as a center of excellence and national focal point for the peaceful applications of space science, remote sensing, and geographic information.In May 2018, Bangladesh launched its first satellite, Bangabandhu-1, into orbit with the help of SpaceX, marking the beginning of the country's space age. This achievement made Bangladesh one of the 57 countries with its own satellite. Additionally, Bangladesh has participated in the NASA Artemis Challenge, which will allow the country to contribute its own Lunar Satellite (LunaSat) to the lunar surface [18]. Bangladesh is one of the 22 countries chosen for this historic project. The teams involved in the project will navigate the challenging phases of design, development, programming, testing, launch, and data collection under the direction of the seasoned professionals of the Great Lunar Expedition for Everyone (GLEE) Program. Also The Bangladesh Delta Plan 2100 (BDP 2100) is a longterm comprehensive strategic and planning initiative developed by the Government of Bangladesh and it involves various stakeholders and collaboration, similar to the space sector [6].

However, political and economic competition between countries can hinder international collaboration, and barriers to international collaboration still exist. Therefore, it is crucial to balance the interests of scientific researchers, the creators of these technologies, and national governments, the users of these technologies, to ensure successful international cooperation in space exploration and usage [8].

B. Regulatory and policy considerations for promoting small satellite launch innovation

Regulatory and policy considerations are crucial for promoting innovation in small satellite launches. The legal and regulatory framework for small satellite missions includes international space law, national legislation, contracts between partners, insurance policies, and the international and national regimes for radio frequency coordination. Compliance with these laws and regulations is essential for planning, launching, and operating a satellite [2]. Failure to comply with regulatory agencies' rules can result in missed launch opportunities, costly system redesigns, and other problems. The regulatory compliance process should play an important role in small satellite mission planning from the earliest stages all the way through launch, operation, and post-mission disposal. The United States Federal Communications Commission (FCC) has recently introduced streamlined processing of satellite license applications for small satellites, reducing application fees and review periods to better accommodate the short timeline between a small satellite's design completion and launch [6]. Space agencies, commercial launch service providers, and other stakeholders must navigate policy compliance and safety requirements for launch and satellite operations. The fastpaced growth of the small satellite industry has presented significant challenges for regulators, who must keep up with

technological advancements and the increasing number of satellites being launched. The growth has raised concerns about space traffic management, the potential for orbital debris, and the need for effective regulatory oversight [20]. Developing globally harmonized regulations and policies, while respecting national interests, remains a significant challenge In conclusion, regulatory compliance is a complicated process that requires careful planning and attention to detail. Small satellite operators must familiarize themselves with the legal and regulatory framework and ensure compliance with the law and other regulatory requirements. Streamlined licensing processes and globally harmonized regulations and policies can help promote innovation in small satellite launch while ensuring safety and sustainability in space exploration and usage [16].

VII. Benefits and potential outcomes

Solving the top three problems in small satellite launch [20], which are space sustainability and safety, cybersecurity and supply chain, and access to space and launch delays, can lead to several benefits and potential outcomes. Small satellites are associated with more rapid development schedules and lower costs. By addressing the challenges in small satellite launch, these desirable characteristics can be further enhanced, allowing for more efficient and cost-effective space missions [11].A key to a successful small satellite program is lowcost, frequent access to space. By solving the challenges in access to space and launch delays, small satellite operators can have more opportunities to deploy their satellites, leading to increased data collection, communication capabilities, and scientific research [8]. Small satellites can provide better views and more accurate measurements of Earth's reflected radiation compared to single, massive satellites. By addressing the challenges in space sustainability and safety, small satellite constellations can be deployed to improve our understanding of Earth's climate, weather patterns, and natural resources [3]. Collaboration and sharing of resources: Small missions can reduce the complexity that comes from many science instruments sharing the same spacecraft and competing for resources. By solving the challenges in cybersecurity and supply chain, small satellite operators can collaborate more effectively, share data and resources, and achieve a higher level of scientific and technological advancement. Small satellite constellations can be used to provide global broadband coverage, enabling reliable and high-speed internet access in remote areas [8] [9]. By addressing the challenges in access to space and launch delays, small satellite operators can deploy these communication networks more quickly and efficiently, benefiting businesses, governments, and individuals worldwide. While most small missions to date have been launched into low Earth orbit [14], addressing the challenges in access to space and launch delays can enable small satellite operators to expand their missions to include space exploration and planetary science interests. This can lead to new discoveries, insights, and advancements in our understanding of the universe.

VIII. Conclusion

Our exploration of strategies and technologies for addressing the challenges of small satellites [20] presents a promising path toward a sustainable space future. Through immersive mission spanning visualization, we can enhance mission planning and preparedness. Small satellite constellations offer improved coverage, reliability, and cost-sharing opportunities. Lunar atmosphere utilization holds the potential to reduce launch mass and enhance sustainability. Meanwhile, the philosophical aspects of future space networks navigate the rapid growth of the small satellite industry [5], collaboration and innovation remain key drivers in our quest to ensure cost-effective and sustainable launches, ultimately shaping a dynamic and responsible space ecosystem.The significance of sustainable and cost-effective small satellite launches cannot be overstated. These launches are pivotal in advancing the small satellite industry, offering numerous advantages.These launches are vital for the industry's growth, innovation, and responsible use of space resources. The outlook for addressing launch challenges in the small satellite industry is crucial. The market is rapidly growing, expected to reach USD 8.86 billion by 2030, driven by technological advances and costeffective launches. However, this growth brings sustainability, safety, and security concerns, requiring collaboration among stakeholders. Collaboration and innovation remain key drivers for the industry's growth and sustainability, as they lead to novel solutions and robust regulations.

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